

Solution Requirements

Date	14 July 2026
Team ID	XXXXXX
Project Name	Personalized Networking Assistant
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

Functional Requirements (FR) define specific behaviors, functions, or features of the system (What the system should do).

Non-Functional Requirements (NFR) specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should be).

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	The MVP does not require formal login; each session is identified by the local browser session for individual/local use during the project scope.	Low
2	Authorization levels	A single "Standard User" role has full access to generate starters, request fact-checks, and view personal history within the session.	Low
3	External interfaces	The system integrates with the Wikipedia Search & Lookup REST API for fact verification of user-requested topics.	High
4	Transactions processing	Each starter-generation and fact-check request is handled as a stateless HTTPS/JSON transaction between the Streamlit frontend and the FastAPI backend, with the result returned immediately to the UI.	High
5	Reporting	The system generates a viewable History report of previously generated starters and their fact-verification status, plus a feedback summary (thumbs up/down counts) per session.	Medium
6	Business rules, etc.	Conversation starters must be generated using both the extracted event themes and any user-supplied interests; feedback must be logged for every generated starter to support future personalization.	High
7	Compliance to laws or regulations	Since bio and interaction data may contain personal information, stored data (User Profiles, Interaction Logs) must follow basic data-privacy good practice and must not be shared externally beyond the Wikipedia lookup call.	Medium
8	Other	Users should be able to clear or export their locally stored history and feedback data on request.	Low

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The DistilBERT + GPT-2 pipeline should return conversation starters quickly enough for interactive use.	Starter generation completes in under 5 seconds per request
2	Scalability	The system should support multiple concurrent local sessions without noticeable performance degradation.	Supports up to 50 concurrent users on a single deployed instance
3	Security & Data Privacy	Locally stored bio data and interaction logs must not be exposed externally.	Data stored only in local JSON files; sent externally only to the Wikipedia API for fact lookups
4	Reliability & Availability	The FastAPI backend and AI/NLP services should remain available throughout a working session.	99% uptime during active local deployment/testing sessions
5	Usability & Accessibility	The Streamlit interface should be simple enough to use without technical training.	Complies with basic WCAG 2.1 AA readability and contrast guidelines
6	Other	Maintainability - the modular backend structure (event_analyzer.py, topic_generator.py, fact_checker.py) should allow independent updates without breaking other components.	Each module independently unit-tested with pytest